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Psychedelics and the Entropic Brain Beyond the Self

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ABSTRACT
A increasing number of studies points toward beneficial effects of psychedelic experiences if administered in the right setting. A smaller number of studies present explanations for why psychedelics have these therapeutic effects. Of these most argue that psychedelic experiences increase neuroplasticity, allowing subjects to let go of unhelpful entrenched beliefs. I argue that (1) psychedelics are likely therapeutic because they help subjects align their beliefs with their respective contexts; (2) relaxation of entrenched beliefs need not occur at higher cognitive levels concerned with the self or ego to be therapeutic; and (3) the proneness toward having supernatural entity experiences can be explained using this contextual approach. I conclude that therapeutic effects of psychedelics are not necessarily tied to beliefs about the self.

Introduction

The past three decades have seen a renaissance into research on the effects of psychedelic substances like psilocybin, LSD and DMT.[1] Recent research converges on the claim that psychedelics are safe and have lasting therapeutic benefits if administered under safe, controlled conditions.[2] While the literature on the therapeutic benefits of psychedelics is vast, most studies agree that psychedelics can have a therapeutic effect. Most are, however, correlational in nature and do not look into the question *why* psychedelics have beneficial effects or how these are brought about. According to a prominent account psychedelics temporarily increase the brain's entropy, which would allow for a therapeutic reset or recalibration. The reset would be therapeutic because it changes subject's beliefs on the self.

In this paper, I argue that therapeutic changes due to psychedelics need not be confined to the self. I survey arguments for the conclusion that the beneficial effects can be explained by a recalibration of information that makes up predictive cognitive models. I argue that such recalibration is likely therapeutic because it allows subjects to fit better with their respective contexts. I also argue that such changes are not solely therapeutic when they impact cognitive models of the self but likely also because they have broader effects. Finally, I discuss how encounters with spiritual or supernatural entities during psychedelic experiences fit within this framework.

This paper is structured as follows. In Section 2, I summarize my main claim and contrast it to other, related claims. In Section 3, I present a brief summary of neuroscientific studies and studies into the phenomenology of psychedelics and argue that they provide support for my main claim. In Section 4, I argue that psychedelically induced recalibrations of cognitive models are therapeutic

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[1]While other substances like mescaline or 2c-b are usually ranked among the psychedelics, fewer studies have looked into their effects. The vast majority of studies focus on psilocybin, LSD or DMT.

[2]See for example: (Grof, 2016).

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when recalibrated models provide a better fit with a subject's context. In Section 5, I discuss how encounters with spiritual or supernatural entities fit within this framework.

Psychedelic experience as contextual recalibration

My aim is to argue that psychedelics allow (1) for a reintegration of information in cognitive models that (2) allows for a better fit with a subject's environment. The first part of my claim is common in the recent debate, be it in a specific form. The second claim is more novel. It differs from other accounts by claiming that the therapeutic effects of psychedelics are not confined to changes in models of the self or ego. Before discussing existing arguments on the effects of psychedelics on cognitive models and my own arguments, I first provide some background.

Background: predictive processing

While considerable questions remain about how human cognition is materialized in the human brain, recent approaches argue that human cognition heavily relies on predictive models of the world encoded in the brain. A number of recent approaches in cognitive neuroscience suggest that humans gradually develop an ever more accurate probabilistic model of their environments.[3] The model generates predictions about sensory input that is compared to actual input picked up by the senses. Any mismatch between both generates friction or entropy which leads to an update of the model of the environment or an action aimed at reducing entropy. The general idea is that the human cognitive system, like most other living systems, strives to reduce entropy by updating its internal model. Revision of the internal model toward greater accuracy is driven by the "free energy principle" (Friston, 2010). The principle describes mathematically how self-organizing agents restrict themselves to a limited number of (likely) sensory encounters by continuously updating expectations about their environment (predictions) and acting in such a way to minimize entropy or surprise. Reducing entropy can occur in two ways: (1) by updating predictions or (2) by optimizing the flow of sensory input (via action). The free-energy principle places an upper bound on entropy or surprise and ensures a degree of certainty or accuracy. The principle requires minimization of an agent's surprise (Demekas et al., 2020).

Processing of sensory input takes a hierarchical nature. Sensory signals are processed bottom-up by a hierarchical model that range from concrete to more abstract representations (Diuk et al., 2013). These are compared to predictions which go in the opposite top-down direction. If the prediction matches the sensory signal, activity at the higher level "cancels out" or "suppresses" the activity at lower levels and the probability of the model is increased. If there is a mismatch, the signals prompt a revision of the model. The revision is implemented at each level. Lower levels in the hierarchy operate on smaller (spatial and temporal) scales and higher levels make predictions at larger scales (Rao & Ballard, 1999). According to Kahl & Kopp, the systems responsible for motor coordination make up the lowest levels of the hierarchy. Sensory signals travel to the level of visual systems next, followed by the sequence level. If required, signals are sent to the motor control network next. At the very top of the hierarchy is the schema level that concerns abstract representations (Kahl & Kopp, 2018). Systems higher up in the hierarchy are located in brain areas that evolved most recently (see Figure 1.)

While human cognitive systems strive toward ever-greater accuracy, certain constraints limit the ability to achieve maximal accuracy. One constraint lies in the computational power of the brain. It is estimated that the human brain's computational power is around 1 exa flop (around 10^{18}) calculations per second (Bostrom & Sandberg, 2008). While the brain's computational power is fast (by comparison, contemporary supercomputers achieve around 1 petaflop (10^{15} calculations per second)), it very likely is unable to represent the world accurately in all its details. Another constraint is efficiency. More

[3]See: (Clark, 2013; Friston, 2010).

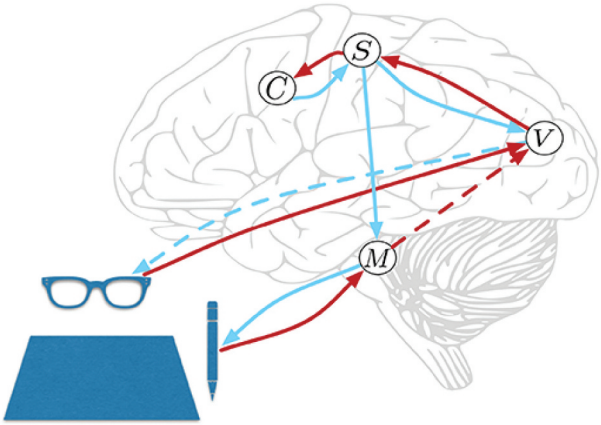

Figure 1. Hierarchical predictive belief (Kahl & Kopp, 2018).

complex, detailed probabilistic models tend to run more slowly and require more time to make calculations. Since humans often need to make quick decisions to survive and flourish, they should not have models that are updated too much or too stringently. For that reason, friction between the internal model and incoming sensory input might often go unnoticed.

Psychedelics as reset

The effects of psychedelics on cognition and subsequently on perception have been described as a "reset" (cf. R. L. Carhart-Harris et al., 2017).[4] Intake of psychedelics would induce a relaxation of precision weighting of prior beliefs. The effects would be most profound when occurring at the highest and deepest levels of the brain's functional architecture. These levels make high-level models like selfhood, identity or ego possible. Relaxation of precision weighting creates a state wherein priors are held with less confidence and are more open to revision. Psychedelic intake thereby increases uncertainty or entropy and opens up the mind's free energy landscape (R. L. Carhart-Harris et al., 2019).

Carhart & Friston compare the relaxation of priors to the act of annealing in computer science and metallurgy. When a system is "heated" it reaches a state of heightened plasticity. The plasticity in turn allows for settling at new energy minima or resting states (R. L. Carhart-Harris et al., 2019). On this model, psychedelics initially increase plasticity and entropy in order to "recalibrate" or reconfigure perceptual models. This in turn would lead to the mind settling on a new and improved perceptual model of the world according to Carhart-Harris and Friston.

The use of adding uncertainty to representation models is widely attested in machine learning. To prevent statistical models from settling on local optima or jumping to quick conclusions (especially in medical diagnostics and image processing) adding uncertainty is beneficial (Hammer & Villmann, 2007). Adding uncertainty through various techniques, like adding noisy data or resampling data, is generally believed to result in better (especially more generalizable) models in the long run.

Carhart- Harris & Friston add that the effects of psychedelics on perception primarily occur at higher levels in the hierarchy. Psychedelic experience would have its most profound effects on higher functions of cognition, like the ego or selfhood, which are rather far removed from immediate sensory processing. Nonetheless the impact on perception would be significant because these higher functions

[4]The term 'reset' is somewhat misleading. Psychedelics (or other practices with similar effects) do not clear away or delete existing predictive models but rather temporarily make them more flexible or malleable. Therefore the term 'annealing' seems more accurate.

contextualize sensory processing, most significantly in the visual modality. The main therapeutic effects of psychedelic experience would also stems from their effects on these higher-up regions (R. L. Carhart-Harris et al., 2019).

The main therapeutic use of psychedelic experience would lie in removing harmful or unhelpful beliefs. Such pathological beliefs can become entrenched in cognitive models. By relaxing precision weighting at higher-level models, these beliefs can be replaced by better alternatives more easily (given the right protocol, set, setting or therapy) (R. L. Carhart-Harris et al., 2019).

Chris Letheby agrees with Carhart-Harris & Friston that the most therapeutically relevant effects of psychedelics are on functions associated with selfhood, identity or ego. He notes that a mere "reset" or "recalibration" of the human biological hardware is not very illuminating on why psychedelics have a profound therapeutic effect. A real explanation for why psychedelics have their therapeutic effects would have to explain what the recalibrated models were doing and how their operations are altered. These features are best explained by changes in (entrenched) beliefs about the self or the ego. Merely relaxing existing models of the self does not suffice either. Recalibration needs to lead to new and improved models of the self. Mainly by affecting the default mode network, psychedelics would relax weighting of priors concerning the self and dissolve pathological ego-beliefs. Examples of pathological ego-beliefs are the belief of not being worthy of anything or a belief of having to be constantly on guard for danger (Letheby, 2021).

Broad recalibration

I noted that a number of prominent accounts attribute the therapeutic effects of psychedelic experience to a relaxation of prior weighting at a high level of the neural hierarchy. The active substance of psychedelics is said to primarily target higher functions of cognition, which in turn lead to top-down induced alterations in perceptual experience and cognition. Some argue that psychedelics mainly target those cognitive functions that give rise to the ego or selfhood and their therapeutic use is found here as well.[5] Sarah Hoffman goes as far as claiming that "it is hard to see how the claim that successful therapy *must* involve changes to the self could be objected to" (Hoffman, 2022, emphasis added). Defenders of this "self-account" tend to suggest that changes to cognitive models concerning the self or ego should be considered broadly. Therapeutic changes (brought about by psychedelics) do not merely include direct alterations to how the self is regarded but also alterations to related models like the boundaries between self and no-self. Nonetheless, much of the changes in cognition brought about by psychedelics (see below) are not therapeutic but of little more than aesthetic or amusement value on this 'self-account'.

I argue that psychedelics likely also therapeutically affect cognition more broadly. By altering various neural functions, psychedelics could trigger new experiences and affect (unconscious) perception,[6] which may in turn prompt a revision of priors on all levels.[7] In doing so, psychedelic experience can remove unhelpful beliefs on multiple levels, including beliefs about the outside world or an alleged transcendent world (see Section 5). On this account, the (therapeutic) effects of psychedelic experience are not limited to higher cognitive functions like the ego. I argue that my account is more in line with the evidence from neuroscience and by reports on the phenomenology of psychedelic experience (see below). My account can also explain the broader therapeutic use of psychedelics for pathologies not associated with selfhood. I argue that the therapeutic effects of

[5]This is also argued by (Pollan, 2021).

[6]Some argue that statements like "unconscious perception" are misnomers because perception necessarily involves awareness (Bennett & Hacker, 2022). I elsewhere argued that this problem can be avoided if perception is defined functionally in the sense of "processing of information" (Van Eyghen, forthcoming). I rely on the same use of "perception" here.

[7]I agree with Letheby and Carhat-Harris & Friston that psychedelic transformation involves increased entropy that can lead to a restructuring or recalibration of cognitive models (see below). Aidan Lyon & Anya Farennikova argue that psychedelic transformation could be brought about by "transparent experiences" (Lyon and Farennikova 2022). My arguments below do not concern this alternative.

psychedelics should be primarily understood from their effects on perceptual experience and to a lesser extent from their effects on unconscious perception. New experiences (whether induced by exogenously or endogenously driven information) can lead to a revision of priors on various levels and thereby lead to a revised (more contextually apt) model of the world. By inducing new experiences psychedelics can help subjects be more in line with their cognitive environment and better cope with the world they live in.

Evidence from cognitive (neuro)science strongly suggests that a significant share of human perception occurs below the level of consciousness. Information that is picked up by the senses without awareness contributes to cognitive models of the world and affects human behavior. Some evidence suggests that psychedelics affect unconsciously processed information as well (see below). Given that unconscious biases or non-reflective reactions can be pathogenic, effects on non-conscious levels of cognition could be therapeutic as well.

In "normal" (non-psychedelic induced) situations, the internal predictive model is built up in a bottom-up fashion. Sensory information is picked up by the lowest levels in the hierarchy. If these prompt a prediction error, the error is transmitted up to higher levels where priors are revised if necessary. A revision of priors in higher levels can in turn lead to revisions at lower levels, which can alter future perceptions. Revision of the internal model is thus normally induced in a bottom-up fashion with a feedback loop by higher levels. Carhart-Harris & Friston and Letheby argue that most therapeutic use of psychedelics is due to their impact on higher cognitive function associated with the self directly.[8] Psychedelics would relax priors at a higher level, which leads to a revised model of the self. Changes at a higher level in turn "contextualize" processing of information at lower levels. I argue that the therapeutic effects of psychedelics are much more like "normal", bottom-up revision of priors. Psychedelics could make the subject attentive to different information. Processing of this information prompts prediction errors, which lead to a revision of the internal model.

Compared to the self-account, my account predicts that changes to cognitive models not associated with the self or ego will sometimes have a therapeutic effect. Having laid out my alternative explanation for the therapeutic use of psychedelics, I will now discuss various studies regarding the neural and phenomenological effects of psychedelics. I will argue that the evidence fits better with my contextual, broad account than it does with the "self-account."

The evidence

In this section, I survey evidence that supports my account. Evidence for the therapeutic effects of psychedelics comes from two broad sources; studies on the effects of psychedelics on various brain functions and studies on the phenomenology of psychedelic experience. The number of studies is vast. My overview is not meant to be exhaustive. I focus on studies that support my view that the therapeutic use of psychedelics stretches beyond their effects on models of the self.

Neuroimaging

As noted, a number of authors argue that psychedelics are therapeutic because they affect higher cognitive functions associated with the self and ego. Neuroscientific studies have shown that psychedelics have effects on other functions not associated with the self or ego as well. This is not denied by defenders of the "self-account," but (tacitly) downplayed as not relevant for their therapeutic effects. Below, I argue that the effects on other cognitive functions likely lead to alterations in experience that allow a subject to build a model of the world that fits better with its context.

[8]Carhart Harris & Friston do argue that psychedelic experience can liberate *bottom-up* information flow via intrinsic sources like the limbic system. They do add that the effects of relaxation of priors is felt most profoundly when it affects the highest or deepest levels of the brain's functional architecture, like those related to self-hood and ego (R. L. Carhart-Harris et al., 2019).

As defenders of the "self-account" note, psychedelics have profound effects on brain regions associated with higher cognitive functions. They point to neural effects in the default mode network and the salience network (encompassing the medial prefrontal cortex, the posterior cingulate cortex/precuneus and the angular gyrus).[9] Psychedelics also affect regions that are not primarily concerned with the self or ego, like the executive control and dorsal attention networks (Tagliazucchi et al., 2014) and thalamus (Timmermann et al., 2019). The executive control network has an important function in integrating sensory and memory information, the regulation of cognition and behavior and participates in working memory (Dong et al., 2020). The dorsal attention network is involved in goal directed control of visuospatial attention (Corbetta & Shulman, 2002). The thalamus has a range of functions among which is coordinating the flow of information from the nervous system to higher regions of the brain (Sherman & Guillery, 2006). Carhart-Harris et al. also noted decreases in activity in the temporal cortex (R. L. Carhart-Harris et al., 2017), which is (among other things) involved in processing sensory auditory information.

Psilocybin intake also affects the claustrum. The functional role of the claustrum has been compared to that of an orchestra conductor. The claustrum integrates information from different regions of the brain and binds information arriving at different times together (Crick & Koch, 2005). Psilocybin intake (10 mg) also decreased the functional connectivity of the right claustrum with auditory and default mode networks, increased right claustrum connectivity with the fronto-parietal task control network and decreased left claustrum connectivity with the fronto-parietal task control network (Barrett et al., 2020). Claustrum activation occurs on the onset of cognitively demanding task and coincides with activation of control networks and the DMN. This is consistent with the idea that the claustrum supports the co-activation of cortical regions that have a functional role for cognitive control of behavior.

Riba et al. noted effects of psychedelics on the limbic system. They note that the affected areas are implicated in emotional arousal (Riba et al., 2006). Timmerman et al also noted effects of psychedelics on the anterior and cingulate cortex in the limbic system (Timmermann et al., 2019). Among other functions, these are involved in emotion formation and processing (Hadland et al., 2003).

Another recurrent finding in MRI- and PET-studies is decreased blood flow and blood-oxygen levels (BOLD) in the cortical networks. A small dose of psilocybin (2 mg) led to changes in BOLD signal spectral behavior which exclusively affects higher brain systems like the default mode, executive control and dorsal attention networks (Tagliazucchi et al., 2014). Other studies confirmed that psilocybin intake led to decreases in blood flow and BOLD signal in the thalamus and anterior and posterior cingulate cortex (ACC and PCC). Intake of psilocybin (10 mg followed by 25 mg, one-week apart) also led to acute decreases in cerebral blood flow in the temporal cortex, including the amygdala with patients suffering from treatment resistant depression (R. L. Carhart-Harris et al., 2017). Another study found consistent but non-significant decreases in the bi-lateral parietal network, the bi-lateral extrastriate cortex network, and bi-lateral frontal cortex network (Muthukumaraswamy et al., 2013).

Another recurring observation is decreased coupling between cortices. Carhart Harris et al. noted a significant decrease in the positive coupling between the medial pre-frontal cortex and posterior cingulate cortex after intake of psilocybin. They argue that the results strongly imply that subjective effects of psychedelic drugs (see below) are brought about by decreased activity and connectivity in the brain's key connector hubs (R. L. Carhart-Harris et al., 2012). The medial pre-frontal cortex is associated with planning, decision making, short-term memory, personality expression, moderating social behavior and controlling certain aspects of speech and language (Yang & Raine, 2009); and the posterior cingulate cortex with awareness, pain and episodic memory retrieval (Nielsen et al., 2005).

Preller et al. noted that intake of LSD led to changes in connectivity between cortico – striato–thalamo-cortical regions. They, however, noted an increase in effective connectivity between the thalamus and the posterior cingulate cortex (Preller et al., 2019). In another study, Preller et al. noted increased sensory-somatomotor brain-wide and thalamic connectivity (Preller et al., 2018). The

[9]For a more complete overview, see: (Letheby 2021: 5.3).

thalamus has several functions, among which relaying sensory information, relaying motor information, regulating attention, awareness and memory (Gazzaniga, 2010).

A small dose of psilocybin (2 mg) was shown to lead to increased functional connectivity between resting state networks, which implies that networks become more closely integrated with one another under the influence of psilocybin. Decreased resting-state functional connectivity was also observed between the visual and sensorimotor resting state networks (Roseman et al., 2014). This may indicate greater independent functioning of these networks.

A final reported effect of psychedelics I note is change in glucose and signal variability. A large dose of psilocybin (15 mg) was found to produce an increase in cerebral metabolic rate of glucose with significant increases in the frontomedial and frontolateral cortex, anterior cingulate cortex and temporomedial cortex. Smaller increases in cerebral metabolic rates of glucose were found in the basal ganglia and even smaller increases in the sensorimotor and occipital cortex (Vollenweider et al., 1997). A dose of 2 mg was shown to lead to increased signal variability in both hippocampi and the anterior cingulate cortex (Tagliazucchi et al., 2014). The frontomedial cortex is involved with numerous sensorimotor, cognitive and affective processes. Examples of these processes are action monitoring, response conflict, reward and decision making.[10]

The short review above strongly suggest that psychedelics have a profound impact on various aspects of cognition. Psychedelics impact brain-areas associated with models of the self or ego, but also a considerable number of other brain-regions. Defenders of the self-account do not deny the broader effects of psychedelics but tend to regard these as either subservient to therapeutic change in models of the self or as therapeutically irrelevant. Given that unhelpful, pathogenic beliefs can cover multiple areas beyond models of the self, such a view is initially unlikely. If psychedelics can (temporarily) increase entropy at various levels of cognition, they can likely lead to therapeutic alterations at various other levels as well. For example, Krebs and Johansen report that lifetime peyote use correlates with significantly lower rates of agoraphobia (Krebs and Johansen 2013); a pathology where subjects fear situations where escape seems difficult. Agoraphobia may have some connections to the self but mainly seems to involve beliefs about the outside world and the danger therein. Given the right input (for example provided by coaching or guidance during a session), psychedelics can therefore likely lead to improved models other than models of the self.

Brain wave patterns

Apart from changes in blood-flow or glucose variability in various brain-regions, psychedelic experience also correlates with profound changes in brain wave frequencies. Different brain wave frequencies are connected to different states of awareness (attention, sleep, waking, etc.). Timmerman et al. argue that reduced alpha and normalized increased delta and theta under DMT may relate to the experience of feeling immersed in an entirely other world. The observed emergence of delta/theta rhythmicity and a sharp reduction in alpha/beta rhythmicity can be linked to the "breakthrough experience," which occurs shortly after DMT intake. During this stage the brain would switch from processing exogenously incoming information to processing endogenously driven information. The same occurs during normal REM sleep dreaming (Timmermann et al., 2019).

Timmerman et al. even go as far as claiming that the characteristic theta- delta rhythmicity under the influence of DMT could have a deep source (e.g., medial temporal lobe) and reflect a very old circuitry that has been associated with REM sleep and medial temporal lobe stimulation. Both are known to feature visionary phenomena. (Timmermann et al., 2019). Alamia et al. conclude that cortical traveling waves and increased forward waves in particular correlate with conscious visual experiences. These could be induced exogenously or endogenously (Alamia et al., 2020).

A dose of DMT (between 0.60 and 0.85 mg per kg body weight) led to decreases in all frequency bands. The decrease was most significant in the theta-, delta- and slow-beta band (Riba et al., 2002).

[10]See: (Moreira et al., 2016) for an overview.

A larger dose of DMT (7–20 mg per kg body weight) correlated with reduces oscillatory power in the alpha- and beta-bands. The initial stage after the DMT intake was also associated with decreases in the delta- and theta-band but both increased after a couple of minutes (Timmermann et al., 2019). Another study confirmed the correlation between DMT intake and decreases in the alpha band. However, while the typical top-down alpha-band rhythms in closed eyes rest were significantly decreased, the bottom-up forward wave was significantly increased (Alamia et al., 2020).

Muthukumaraswamy et al. note reduced spontaneous cortical oscillatory power from 1 to 50 Hz in posterior association cortices and from 8 to 100 Hz in frontal association cortices. They also note large decreases in oscillatory power in the default-mode network but no notable effect on low-level visually induced and motor-induced gamma band oscillations. They conclude that the subjective effects of psychedelics are likely the result from a desynchronization of ongoing oscillatory rhythms in the cortex, likely triggered by 5-HT_{2A} receptor-mediated excitation of deep pyramidal cells (Muthukumaraswamy et al., 2013).

According to Alamia et al. decreases in top-down alpha-band and increases in bottom-up forward wave support the idea that psychedelics reduce the precision weighting of priors by altering the balance between top-down versus bottom-up information passing (Alamia et al., 2020). Carhart-Harris and Karl Friston make a similar claim (R. L. Carhart-Harris et al., 2019).

To my knowledge, changes in neural oscillation do not have direct effects on cognitive models of the self or ego. The therapeutic use of fostering theta- and alpha brainwaves is well attested. Apart from aiding in finding a balanced connection between the inner world of the self and the outer world of objects, alpha waves can also foster a state of sereneness and peacefulness (White & Richards, 2009). Theta waves also contribute to mental relaxation. Psychedelic experience could therefore contribute to relaxation and states of lower stress without the need for an effect on models of the self. Since relaxation and stress reduction are obviously therapeutic, they can thus likely be therapeutic without impacting models of the self or ego.

Phenomenology of psychedelic experience

MRI-, PET- and EEG-studies strongly suggest that psychedelics have a significant impact on brain networks and functions that modulate various aspects of human cognition. Additional support for a broad impact on human cognition comes from studies into the phenomenological nature of psychedelic experiences. Psychedelic experiences are notoriously variable and subject to influences of setting, context, and individual differences. Nonetheless a couple of recurring features or patterns can be distilled, among which are changes in perception and processing of information. While some of the effects address aspects of personality or the ego, this is not always the case. Cases where subjects report a change of outlook about the outside worlds are perceived as therapeutic as well.

Most studies on the phenomenology of psychedelic experience focus on DMT or psilocybin. According to Preller and Vollenweider the psychological peak effects of psilocybin cannot be distinguished from those of LSD or mescaline in controlled studies. Subjects do ascribe different emotional salience to different psychedelics (Preller & Vollenweider, 2016). For example, some describe the experience of mescaline as “gentler” or less abrupt’ than that of psilocybin. The differences in perception or cognition appear to be similar.

Subjects administered small doses of DMT (0,04 to 0,4 mg per kg body weight) reported visual hallucinations (with eyes opened or closed), including both familiar and novel images, and bodily dissociation. A recurring feature was perception of geometric figures or decomposition of perceptual input into more basic geometric shapes. Some also reported perception of greater depth, seeing reality as “much more than 3-dimensional” (Strassman et al., 1994). In another study, subjects reported a “kaleidoscopic display” of intensely colored abstract and representational images (Strassman, 1995).[11] Some participants described DMT-induced experiences as similar to dreams (Strassman

[11]See also: (Díaz, 2010).

et al., 1994), others as more real and compelling than ordinary experience (Strassman, 1995). Some participants in experiments reported changes in auditory effects as well. Usually, they heard high-pitched sounds resembling whining, chattering or crackling noises (Strassman et al., 1994).

The studies thus strongly suggest that psychedelics have a profound effect on subject's perception and processing of sensory information. It includes alteration in different sense modalities and in the perceived nature of reality.

Roseman et al. report that the most salient factors contributing to overall well-being two weeks after psychedelic experience are resolution of a personal conflict or trauma, emotional breakthroughs and feelings of closure on an emotional problem (Roseman et al., 2019). Barrett et al. note a strong correlation between experiences of ego-dissolution (see also below) and well-being, but also note effects of spiritual significance on well-being (Barrett et al., 2020).

Some studies note changes in the perception of the self. Preller and Vollenweider note that subjects who took psilocybin often experience a "pleasurable state of self-dissolution," alterations of ego-boundaries or a sense of unity. Sometimes the experiences can go as far as loss of control over one's own actions or loss of thought control. Chris Letheby surveys a number of self-reported experiences wherein subjects report changes in self-perception as therapeutic (Letheby, 2021: chapter 7). Subjects also report less intrusive experiences like altered sense of time, positively heightened mood, mania like experiences, visual hallucinations, auditory alterations, hypersensitivity, synesthesias or vivid recollection of memories (Preller & Vollenweider, 2016).

Anecdotal evidence also supports therapeutic effects of experiences not obviously related to the self. A recurring effect of psychedelic experience is diminished fear of death. An anonymous subject writes after meeting various spirits during an ayahuasca-trip:

> I feel like ayahuasca prepares you for death as well. I am no longer afraid of death. I will all eventually die. It takes away the mystery; allows you to at least imagine what the afterlife might be like. What's the point of believing nothing happens after death anyway? It's fucking depressing .[12]

Some more recent evidence also appears to support my claim that psychedelics can have a therapeutic use without making alterations to models of the self or ego. Olson argues that psychedelics need not even have any subjective (phenomenal) effects to be therapeutic. He surveys evidence that microdosing, which has far less notable phenomenal effects, appears to lead to greater neuroplasticity compared to no intake. Microdosing would also have lasting beneficial effects (Olson, 2020). Others insist on the need for subjective experiences but include changes that do not immediately pertain to models of the self like positive mood and positive behaviors (Yaden & Griffiths, 2020). Patients suffering from depression and anxiety report positive changes in life, mood, relationships and spirituality apart from improved attitudes toward the self (Griffiths et al., 2016). Carbonero et al. point to the ability of psilocybin to produce experiences of visual effects, positive mood, insight, positive social effects, increased awareness of beauty (both visual and music), awe or amazement, meaningfulness, and mystical experience (Carbonaro et al., 2020). Of these, only positive mood, insight and meaningfulness seem closely related to models of the self in many cases. Positive social effects at least partially pertain to models on others; increased awareness of beauty and awe[13] appear to pertain foremost to the external world; and mystical experience frequently involves what is regarded as transcendent (see below). The evidence suggests that changes on lower levels of cognitive models concerned with emotional arousal and immediate apprehension of the environment can have therapeutic effects as well.

[12]https://erowid.org/experiences/exp.php?ID=91566.

[13]Letheby does connect awe to changes in cognitive models of the self (Letheby, 2021: 112). However, while experiences of awe can trigger an increased sense of humility or connectivity, they need not always do so and can remain a sense of what objects in the external world are like.

Wrap up

In summary, whereas Carhart-Harris and Firston and Letheby strongly suggest that the therapeutic effects of psychedelics are mainly or even exclusively due to their effects on higher cognitive functions like the self, I argue that relaxation of priors at lower cognitive levels and subsequent convergence on new cognitive models can have profound therapeutic effects as well. I reviewed evidence that the therapeutic effects of psychedelics are likely broader, affecting all kinds of experiences and unconscious cognition.

While models of the self are undoubtedly involved in various kinds of cognition, questions remain about its scope. It is unlikely that models of the self are involved in all instances of common ordinary cognition, especially at lower levels. If they do, the self would become all-encompassing and any distinction between cognitive models of the self and models of the external world would become blurred. While all cognition can be plausibly related to a "minimal self," that is the carrier of all experience and cognition, claiming that all cognition is therefore about models of the self risks rendering any self-account trivial. A similar problem arises when defenders of a self-account related all aspects of meaningfulness to models of the self. While some aspects of meaningfulness (like a sense of unity or connectivity during mystical experiences) are plausibly related to models of the self, other aspects like a changed view toward the world or the transcendent (see Section 5) are not.

Recalibration as contextual

Having explained my model and how it differs from alternate models, I now move to a discussion of what makes a cognitive model therapeutically *better*. Studies and anecdotal evidence support the idea that while psychedelic intake can have therapeutic effects, it can likewise have nefarious effects, especially when administered in high doses (cf. Barrett et al., 2020). I argue that a cognitive model (whether concerning the self or not) is therapeutically better when it is more in line with the subject's context.[14]

I noted that psychedelics have a profound impact on the brain's cognitive functions according to Carhart-Harris and Friston. The effects would result in relaxation of priors. After or during psychedelic experiences, priors are again "hardened" and the subject converges toward new models and new beliefs. One potential reason why new models and beliefs would be therapeutically better is epistemic in nature. New models could be more in line with the real world and therefore allow a subject to navigate her environment with less friction.[15] While an epistemic improvement can indeed be therapeutic, I argue that non-epistemic improvements may be therapeutic as well. Letheby argues that relaxation of priors due to psychedelics allows for epistemic benefits, like epistemically superior cognitive models, yet also provides an avenue for epistemic harm (Letheby, 2021).

The therapeutic effect of psychedelic experiences can also be explained differently. Psychedelic experiences can have multiple (incompatible) long-term effects on people. For example, Lutkajtis notes that psychedelic experiences often lead to a renewed interest in religion. This would follow from alleged encounters with supernatural beings during psychedelic experiences (Lutkajtis, 2020). Carhart-Harris and Friston and Chris Letheby note that psychedelic experiences need not involve any experience beyond the natural world (Letheby, 2021; R. L. Carhart-Harris et al., 2019). In experiences of the first kind, psychedelic experiences likely lead to a revision of cognitive models toward more openness for supernatural beliefs; in experiences of the second kind it does not. Psychedelic experiences can thus likely lead to revision in multiple directions.

[14]Joshua M. Martin and Philipp Sterzer argue for the vital role of context. Unlike mine, their account focuses on the role of context in the *discovery* of new cognitive models of the self (Martin and Sterzer 2022). I argue that fitting with context is that what makes improved cognitive models therapeutic.

[15]While Carhart-Harris appear to argue that models are improved when brought in line with a subject's context, they suggest an epistemic function as well. For example, they write: "a commitment to the discovery of better models is important, as improved models will provide more robust and reliable approximations to reality, which are likely to support greater stability in terms of mental health" (R. L. Carhart-Harris et al., 2019, p. 335).

Revision or recalibration of cognitive models after psychedelic experiences thus not always leads toward epistemically superior models of the world. I propose that epistemic revision of models is only of secondary importance in understanding the therapeutic use of psychedelics. In order to reduce surprise, cognitive models need not always be more accurate. What matters is that models allow subjects to make sense of their environments and are workable in the contexts wherein subjects find themselves. For some subjects, a religious outlook will allow them to make sense of more sensory stimuli whereas for others a religious outlook will imply an overburdened view of the world.

Understanding the therapeutic use of psychedelics as helping subjects align cognitive models to their present contexts fits well with the importance of setting and preparation in psychedelic therapies. Facilitators of psychedelic therapies (whether clinical or not) often stress the need for thorough preparation and sometimes of voicing the problem or need on which the subject needs help or guidance.[16] Setting a positive mood would lower the risk of negative experiences after psychedelic intake. Given that such negative experience could plausibly produce unhelpful beliefs or trauma, lowering the risk is likely conducive to cognitive models more in line with a subject's context. They also stress the need for continued reflection and integration after psychedelic experience (e.g., Blewett & Chwelos, 1959). Continued after-effects of psychedelics suggest that recalibration of cognitive models after relaxation by psychedelics can take up some time. New experiences may take time to interpret properly or to see their meaning. Without proper preparation or integration psychedelic experiences may not be therapeutic or even dangerous.[17]

Some reported similar effects on cognition from meditation practices. Laukkonen and Slagter propose that meditation affects all levels of the predictive hierarchy in virtue of changes in the deployment and relaxation of attention (Laukkonen & Slagter, 2021). Other practices like states of relaxation (Fox et al., 2013), shamanic trances and drumming (Gingras et al., 2014; Neher, 1962; Oohashi et al., 2002) or even epileptic discharges (Danielson et al., 2011) have similar effects on brain wave oscillation to intake of psychedelics (Marsh, 2021). These and other practices may therefore also lead to relaxation of priors and allow for recalibration. Apart from some forms of meditation these other practices are rarely used in therapeutic settings. Other therapeutic practices explicitly have the goal of revising entrenched unhelpful beliefs. For example, during Eye Movement Desensitization and Reprocessing (EMDR) Therapy patients are encouraged to briefly focus on the trauma memory while simultaneously experiencing bilateral stimulation (typically eye movements). This is associated with a reduction in the vividness and emotion associated with the trauma memories (American Psychological Association, 2017). EMDR-therapy is more focused than a psychedelic experience and therefore likely better suited to cope with specific traumatic memories. Nonetheless, EMDR-therapy and psychedelic experience appear to have related effects. EMDR-therapy also does not aim at increased epistemic accuracy but rather at an updated view of the world that is better suited for the subject's current context.

Recontextualization and entity encounters

Some argued that the therapeutic effects of psychedelics are connected to their enabling mystical or supernatural experiences (e.g., Lutkajtis, 2020). Others argue that naturalistic experiences can be therapeutic as well (e.g., Letheby, 2021). While in any case psychedelic experience does not necessarily involve encounters with spiritual or supernatural entities and such encounters are not required for their therapeutic use, they often do. Some psychedelic experiences involve encounters with seemingly autonomous entities that seem to possess intelligence and agency (Lutkajtis, 2020). Others involve more general mystical experiences like feelings of unity or oneness, transcendence of time and space,

[16]The unofficial handbook for LSD-therapy by Blewett and Chwelos stresses the need for a quiet setting free from disturbing noises and good rapport between therapist and patient. The authors also encourage subjects to carefully reflect on their main problems before the sessions (Blewett & Chwelos, 1959).

[17]See (Carhart-Harris, 2013) for a discussion of possible malignant effects of psychedelic experience.

deeply felt positive moods, sense of awe, sense of reverence and wonder and ineffability (Lutkajtis, 2020; Roseman et al., 2018). Below I argue that my contextual approach can explain why some have supernatural encounters during psychedelic experience and others do not. I also argue that such experiences can be therapeutic and provide explanations as to why.

The fact that psychedelics can foster experiences in line with both a supernatural or theistic and naturalistic worldview shows that psychedelics can direct experiences in different, mutually opposing ways.[18] I argue that the direction in which psychedelics guide the subject are sometimes a function of the specific needs the subjects has in her context. This supports my claim that psychedelic experience is primarily therapeutic because it allows for recalibration befitting to the subject's context rather than epistemic recalibration.

Some argue that the prevalence of religious experiences is explained by differences in prior religious beliefs. Marc Andersen argues that people with prior beliefs in supernatural beings will have a stronger disposition to have religious or mystical experiences. Because their inner cognitive model of the world contains supernatural beings like God or spirits, religious believers will be more prone to see earthly phenomena as caused by a supernatural being. Subjects with a more naturalistic model will be less prone to have similar experiences (Andersen, 2019). Applied to psychedelic experience, religious believers would be more inclined to encounter supernatural beings than skeptics because they have priors that refer to such beings.

While Andersen's account can explain much of the distribution of supernatural and natural psychedelic experience, it has problems explaining psychedelic experiences wherein non-religious subjects encounter spiritual beings or psychedelic experiences that culminate in conversions. In one study, more than two thirds of subjects identifying as "atheist" no longer did so after the experience (Griffiths et al., 2019).[19] A different study noted a drop in self-reported atheism (from 28% to 10%) and increase in self-reported identification as "belief in ultimate reality, higher power, God, or universal divinity" (36% to 58%) after DMT-induced experiences (Davis et al., 2020).

Apart from epistemic questions,[20] encounters with supernatural beings during psychedelic experience also raise questions about their therapeutic role. Is there reason to think that encounters with spiritual entities during psychedelic experience can be therapeutic? Online blogs and repositories provide ample anecdotal evidence for this claim.[21] Davis et al. note that although some experience encounters with spiritual entities as fearful, a large share rated these experiences as "among the most meaningful, spiritual, and psychologically insightful lifetime experiences, with persisting positive changes in life satisfaction, purpose, and meaning attributed to the experiences." (Davis et al., 2020, 1008). Podrebarac et al. report of an experience during psychedelic therapy:

> [Joan] opened her eyes and reported to her guides that she saw her deceased mother present in the room with her standing to her left. Because contact with her mother was something she had long hoped to do, the guides encouraged Joan to make inquiry, to feel, and to communicate with her mother' s presence. Joan then entered into dialogue with her mother. As she asked her mother questions aloud, Joan would then close her eyes and hear her mother' s response from within. She understood this communication to be coming from her mother, whom she now understood to be very much connected with her. In this communication, Joan received reassurance that her mother was with her always and that her mother loved her deeply. She experienced this as deeply restorative and affirming (Dobrevac: 181).

A final remaining question is why encounters with spiritual entities can be experienced as therapeutic. There is ample literature on religious coping, i.e. "religiously framed cognitive, emotional, or behavioral responses to stress, encompassing multiple methods and purposes as well as positive and

[18]One could argue that some naturalistic experiences are compatible with a supernaturalisc worldview. This, however, does not apply to all naturalistic experiences.

[19]Matthew Johnson and David Yaden warn against drawing strong conclusions about the effects of psychedelics on religiosity (Johnson & David, 2020). The evidence does appear to suffice to show that responsivity to supernatural encounters is not determined by prior religious beliefs.

[20]See for example: (Letheby, 2017, 2021).

[21]See for example: https://www.samwoolfe.com/2020/12/overcoming-the-fear-of-death-during-a-psychedelic-experience.html.

negative dimensions" (Gellman et al., 2013). A considerable number of studies point to beneficial effects of various aspects of religiosity (church attendance, petitionary prayer, religious beliefs, etc.) on health (e.g., Koenig & Cohen, 2002). Some, however, point out that the relation between religiosity and health is ambiguous.[22]

A considerable number of subjects report support or guidance from supernatural entities.[23] Part of the distribution of natural and supernatural experiences can be explained by their contribution to contextual recalibration. Anna Lutkajtis notes that entities encountered during psychedelic experience are often felt to be powerful, wise and loving (Lutkajtis, 2020). David Luke notes that such encounters are frequently accompanied by positive emotions like joy, trust, love, kindness and friendship (Luke, 2017). John Heuser concluded from a thematic analysis of entity encounters after ayahuasca intake that entities frequently expressed human emotions of love and concern for human welfare. Entities often take on the role of a guide and guide subjects toward clarification and integration after contemplating difficult problems or dark themes. Furthermore, entities sometimes help subjects to engage in 'biographical recapitulation, where scenes from the subject's autobiographical past were relived or revisited (Heuser, 2006). Another way in which entities help subjects is by providing insightful information about the subjects themselves and the world they live in (Luke, 2017).

A next way in which entities can be of aid is by inducing experiences of awe. For example, A. Khan recounts the following experience:

> The huge portion of entity that remained continued to gaze down upon me and I could feel the terrific weight of its otherworldly stare. I felt there was simply nothing I could do that would properly convey my recognition, my respect and awe for that which had just revealed itself to me (Khan, n.d.).

As Lutkajtis notes, the feeling evoked by being in the presence of something awe-inspiring that transcends current frames of reference – can have beneficial effects on wellbeing (Lutkajtis, 2020). While experiences of awe often include a sense of being small, which connects to models of the self, it also involves an altered sense of what the external world is like as the quote above illustrates.

Finally, entity encounters can help subjects cope with difficulties from the past or trauma. Tramacchi notes that a considerable number of DMT accounts have elements of "shamanic ordeal" (Tramacchi, 2006). During such ordeals, subjects go through troubling episodes that include pain, stress or even the sense of dying. Such experiences can have a cleansing effects whereby the subject is freed from negative emotions and traumatic memories.[24]

The overview explains how entities encountered during psychedelic experiences very often have a therapeutic role. They aid subjects in processing autobiographical memories or guide subjects toward life-altering experiences. Such experiences fit well the claim that psychedelic experiences are therapeutic because they enable recalibration of (deeply entrenched) priors. My account where recalibration is therapeutic because it helps the subject develop priors that fit well with her environment can help explain the distribution of supernatural entities encounters. For some personalized encounters will be better suited to revise weighted priors. This might be because they prefer personalized interactions in daily life.[25] It might also be because personalized encounters are better suited in some situations.

A contextualized approach toward the therapeutic effects of recalibration of priors can therefore explain why some people have encounters with personal supernatural entities during psychedelic

[22]See: (Szocik & Van Eyghen, 2021: chapter 4) for a discussion.

[23]This paragraph draws heavily on Anna Lutkajtis's overview (Lutkajtis, 2020).

[24]A defender of the self-account could argue that freeing from traumatic memories is an instance of a changed cognitive model of the self. Such a reply assumes that trauma is part of personality of selfhood. While a proper assessment of this claim lies beyond the scope of this paper it is not obviously true.

[25]This might be connected with differences in the operations of the theory of mind. The Theory of Mind (ToM) allows humans to make sense of other people's behavior by forming beliefs about other people's mental states. People with a more developed ToM are generally better at interacting with others. People with deficits (e.g., those suffering from autism spectrum disorders or attention deficit disorders) have more difficulties in doing so. It is possible that subjects with less developed ToM abilities are less inclined to have encounters with entities during psychedelic experience. More research is, however, needed on this topic.

experiences and others have more general mystical experiences or mere natural experiences. While such experiences might be helpful or therapeutic for some, they might not be for others.

The therapeutic effects of psychedelic encounters with supernatural entities also provides additional support against the self-account. While some encounters have effects on the self, e.g., relief of trauma, others mainly result in an altered view of the universe or ultimate reality. These are not obviously connected to the self, yet therapeutic.

Conclusion

Psychedelics will likely be of continued interest to psychology (of religion) for various reasons. I argued that:

(1) psychedelics have a therapeutic effect because they allow for changes in cognitive models and beliefs at various levels (including but not limited to models of the self or ego);
(2) psychedelics likely have their therapeutic effects because they allow for cognitive models more attuned to a subject's context;
(3) encounters with spiritual entities during psychedelic experiences may be therapeutic and discussed reasons why.

While I surveyed evidence in support of all three claims (like neural effects unconnected to areas involved with the self and reports of therapeutic experiences unconnected to the self), the evidence is often anecdotal and limited. My main goal was providing an account which I deem more plausible than the self-account, which is in line with the available evidence. More research into the effects of psychedelics on cognitive models not involved with the self or ego can provide extra support or discard my account.

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References

Alamia, A., Timmermann, C., Nutt, D. J., VanRullen, R., & Carhart-Harris, R. L. (2020). DMT alters cortical travelling waves. *Elife*, *9*, e59784. https://doi.org/10.7554/eLife.59784

American Psychological Association. 2017. Eye movement desensitization and reprocessing (EMDR) therapy. https://www.apa.org/ptsd-guideline/treatments/eye-movement-reprocessing.

Andersen, M. (2019). Predictive coding in agency detection. *Religion, Brain & Behavior*, *9*(1), 65–84. https://doi.org/10.1080/2153599X.2017.1387170

Barrett, F. S., Krimmel, S. R., Griffiths, R. R., Seminowicz, D. A., & Mathur, B. N. (2020). Psilocybin acutely alters the functional connectivity of the claustrum with brain networks that support perception, memory, and attention. *NeuroImage*, *218*, 116980. https://doi.org/10.1016/j.neuroimage.2020.116980

Bennett, M. R., & Michael Stephan Hacker, P. (2022). *Philosophical foundations of neuroscience*. John Wiley & Sons.

Blewett, D. B., & Chwelos, N. (1959). Handbook for the therapeutic use of lysergic acid diethylamide-25 individual and group procedures. Retrieved October 14, 2022, from https://maps.org/research-archive/ritesofpassage/lsdhandbook.pdf

Bostrom, N., & Sandberg, A. (2008). Whole brain emulation: A roadmap. Lancaster University. Retrieved September 05, 2022, from https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf

Carbonaro, T. M., Johnson, M. W., & Griffiths, R. R. (2020). Subjective features of the psilocybin experience that may account for its self-administration by humans: A double-blind comparison of psilocybin and dextromethorphan. *Psychopharmacology*, *237*(8), 2293–2304. https://doi.org/10.1007/s00213-020-05533-9

Carhart-Harris, R. (2013). Psychedelic drugs, magical thinking and psychosis. *Journal of Neurology, Neurosurgery & Psychiatry*, *84*(9), e1–e1. https://doi.org/10.1136/jnnp-2013-306103.17

Carhart-Harris, R. L., Erritzoe, D., Williams, T., Stone, J. M., Reed, L. J., Colasanti, A., & Tyacke, R. J., Leech, R., Malizia, A. L., Murphy, K. and Hobden, P. 2012. Neural correlates of the psychedelic state as determined by FMRI studies with psilocybin. *Proceedings of the National Academy of Sciences* 109 (6), 2138–2143.

Carhart-Harris, R. L., Friston, K. J., & Barker, E. L. (2019). REBUS and the anarchic brain: Toward a unified model of the brain action of psychedelics. *Pharmacological Reviews*, *71*(3), 316–344. https://doi.org/10.1124/pr.118.017160

Carhart-Harris, R. L., Roseman, L., Bolstridge, M., Lysia Demetriou, J. N. P., Wall, M. B., Tanner, M., Tanner, M., Kaelen, M., McGonigle, J., Murphy, K., Leech, R., Curran, H. V., & Nutt, D. J. (2017). Psilocybin for treatment-resistant depression: FMRI-Measured brain mechanisms. *Scientific Reports*, *7*(1), 1–11. https://doi.org/10.1038/s41598-017-13282-7

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *The Behavioral and Brain Sciences*, *36*(3), 181–204. https://doi.org/10.1017/S0140525X12000477

Corbetta, M., & Shulman, G. L. (2002). Control of goal-directed and stimulus-driven attention in the brain. *Nature Reviews Neuroscience*, *3*(3), 201–215. https://doi.org/10.1038/nrn755

Crick, F. C., & Koch, C. (2005). What is the function of the claustrum? *Philosophical Transactions of the Royal Society B: Biological Sciences*, *360*(1458), 1271–1279. https://doi.org/10.1098/rstb.2005.1661

Danielson, N. B., Guo, J. N., & Blumenfeld, H. (2011). The default mode network and altered consciousness in epilepsy. *Behavioural Neurology*, *24*(1), 55–65. https://doi.org/10.1155/2011/912720

Davis, A. K., Clifton, J. M., Weaver, E. G., Hurwitz, E. S., Johnson, M. W., & Griffiths, R. R. (2020). Survey of entity encounter experiences occasioned by inhaled N, N-Dimethyltryptamine: phenomenology, interpretation, and enduring effects. *Journal of Psychopharmacology*, *34*(9), 1008–1020. https://doi.org/10.1177/0269881120916143

Demekas, D., Parr, T., & Friston, K. J. (2020). An investigation of the free energy principle for emotion recognition. *Frontiers in Computational Neuroscience*, *14*, 30. https://doi.org/10.3389/fncom.2020.00030

Díaz, J. L. (2010). Sacred plants and visionary consciousness. *Phenomenology and the Cognitive Sciences*, *9*(2), 159–170. https://doi.org/10.1007/s11097-010-9157-z

Diuk, C., Tsai, K., Wallis, J., Botvinick, M., & Niv, Y. (2013). Hierarchical learning induces two simultaneous, but separable, prediction errors in human Basal Ganglia. *Journal of Neuroscience*, *33*(13), 5797–5805. https://doi.org/10.1523/JNEUROSCI.5445-12.2013

Dong, W., Qiu, C., Jiang, X., Shen, B., Zhang, L., Liu, W., Zhang, W., & Chen, J. (2020). Can the executive control network be used to diagnose Parkinson's Disease and as an efficacy indicator of deep brain stimulation? *Parkinson's Disease*, *2020*, 1–6. https://doi.org/10.1155/2020/6348102

Fox, K. C., Nijeboer, S., Elizaveta Solomonova, G. W. D., & Christoff, K. (2013). Dreaming as mind wandering: Evidence from functional neuroimaging and first-person content reports. *Frontiers in Human Neuroscience*, *7*, 412. https://doi.org/10.3389/fnhum.2013.00412

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, *11*(2), 127–138. https://doi.org/10.1038/nrn2787

Garcia-Romeu, A. P., & Tart, C. T. (2013). Altered states of consciousness and transpersonal psychology.

Gazzaniga, M. S. (2010). *The cognitive neuroscience of mind: A tribute to Michael S. Gazzaniga*. MIT Press.

Gellman, M. D., Turner, R., & Wortmann, J. (2013). Religious coping. In M. D. Gellman & J. R. Turner (Eds.), *Encyclopedia of behavioral medicine*. Springer.

Gingras, B., Pohler, G., Tecumseh Fitch, W., & Kotz, S. (2014). Exploring shamanic journeying: Repetitive drumming with shamanic instructions induces specific subjective experiences but no larger cortisol decrease than instrumental meditation music. *Plos One*, *9*(7), e102103. https://doi.org/10.1371/journal.pone.0102103

Griffiths, R. R., Hurwitz, E. S., Davis, A. K., Johnson, M. W., & Jesse, R. (2019). Survey of subjective" God Encounter Experiences": Comparisons among naturally occurring experiences and those occasioned by the classic psychedelics Psilocybin, LSD, Ayahuasca, or DMT. *Plos One*, *14*(4), e0214377. https://doi.org/10.1371/journal.pone.0214377

Griffiths, R. R., Johnson, M. W., Carducci, M. A., Umbricht, A., Richards, W. A., Richards, B. D., Cosimano, M. P., & Klinedinst, M. A. (2016). Psilocybin produces substantial and sustained decreases in depression and anxiety in patients with life-threatening cancer: A randomized double-blind trial. *Journal of Psychopharmacology*, *30*(12), 1181–1197. https://doi.org/10.1177/0269881116675513

Grof, S. (2016). *Realms of the human unconscious: Observations from LSD research*. Souvenir Press.

Hadland, K. A., Rushworth, M. F., Gaffan, D., & Passingham, R. E. (2003). The effect of cingulate lesions on social behaviour and emotion. *Neuropsychologia*, *41*(8), 919–931. https://doi.org/10.1016/S0028-3932(02)00325-1

Hammer, B., & Villmann, T. (2007). *European Symposium of Artificial Neural Networks* (pp. 79–90).

Heuser, J. 2006. Ayahuasca entity visitations: A thematic analysis of internet-reported encounters. California Institute of Integral Studies.

Hoffman, S. (2022 November, 5). Positive affect and Letheby's naturalization of psychedelic therapy. *Philosophy and the Mind Sciences*, *3*. https://doi.org/10.33735/phimisci.2022.9285

Johnson, M., & David, Y. (2020). There's no good evidence that psychedelics can change your politics or religion by Matthew W. Johnson, David B. Yaden on November 5, 2020. *Scientific American*.

Kahl, S., & Kopp, S. (2018). A predictive processing model of perception and action for self-other distinction. *Frontiers in Psychology*, *9*, 2421. https://doi.org/10.3389/fpsyg.2018.02421

Khan, D. (n.d.). *DMT & my occult mind II: Investigation of occult realities using the spirit molecule (Book One)*. CreateSpace Independent Publishing Platform.

Koenig, H. G., & Cohen, H. J. (2002). *The link between religion and health*. Oxford University Press.

Krebs, T. S., & Johansen, P.Ø. (2013). Psychedelics and mental health: A population study. *Plos One*, *8*(8), e63972.
Laukkonen, R. E., & Slagter, H. A. (2021). From many to (n) one: Meditation and the plasticity of the predictive mind. *Neuroscience & Biobehavioral Reviews*, *128*, 199–217. https://doi.org/10.1016/j.neubiorev.2021.06.021
Letheby, C. (2017). Naturalizing psychedelic spirituality. *Zygon®*, *52*(3), 623–642. https://doi.org/10.1111/zygo.12353
Letheby, C. (2021). *Philosophy of Psychedelics*. Oxford University Press.
Luke, D. (2017). *Otherworlds: Psychedelics and exceptional human experience*. Muswell Hill Press.
Lutkajtis, A. (2020). Entity encounters and the therapeutic effect of the psychedelic mystical experience. *Journal of Psychedelic Studies*, *4*(3), 191–198. https://doi.org/10.1556/2054.2020.00143
Lyon, A., & Farennikova, A. (2022). Through the psychedelic looking-glass: The importance of phenomenal transparency in psychedelic transformation. *Philosophy and the Mind Sciences*, *3*.
Marsh, M. (2021). *Sensing the divine influences of near-death, out-of-body & cognate neurology in shaping early religious behaviours*. Springer.
Martin, J. M., & Sterzer, P. (2022). How level is the'cognitive playing field'? Context shapes alterations in self-conception during the psychedelic experience. *Philosophy and the Mind Sciences*, *3*.
Moreira, P. S., Marques, P., & Magalhães, R. (2016). Identifying functional subdivisions in the medial frontal cortex. *Journal of Neuroscience*, *36*(44), 11168–11170. https://doi.org/10.1523/JNEUROSCI.2584-16.2016
Muthukumaraswamy, S. D., Carhart-Harris, R. L., Moran, R. J., Brookes, M. J., Williams, T. M., Errtizoe, D., Sessa, B., Papadopoulos, A., Bolstridge, M., Singh, K. D., Feilding, A., Friston, K. J., & Nutt, D. J. (2013). Broadband cortical desynchronization underlies the human psychedelic state. *Journal of Neuroscience*, *33*(38), 15171–15183. https://doi.org/10.1523/JNEUROSCI.2063-13.2013
Neher, A. (1962). A physiological explanation of unusual behavior in ceremonies involving drums. *Human Biology*, *34*(2), 151–160.
Nielsen, F. Å., Balslev, D., & Kai Hansen, L. (2005). Mining the posterior cingulate: Segregation between memory and pain components. *Neuroimage*, *27*(3), 520–532. https://doi.org/10.1016/j.neuroimage.2005.04.034
Olson, D. E. (2020). The subjective effects of psychedelics may not be necessary for their enduring therapeutic effects. *ACS Pharmacology & Translational Science*, *4*(2), 563–567. https://doi.org/10.1021/acsptsci.0c00192
Oohashi, T., Kawai, N., Honda, M., Nakamura, S., Morimoto, M., Nishina, E., & Maekawa, T. (2002). Electroencephalographic measurement of possession trance in the field. *Clinical Neurophysiology*, *113*(3), 435–445. https://doi.org/10.1016/S1388-2457(02)00002-0
Pollan, M. (2021). *This is your mind on plants*. Penguin.
Preller, K. H., Burt, J. B., Jie Lisa, J., Schleifer, C. H., Adkinson, B. D., Stämpfli, P., Seifritz, E., Repovs, G., Krystal, J. H., Murray, J. D., Vollenweider, F. X., & Anticevic, A. (2018). Changes in global and thalamic brain connectivity in LSD-Induced altered states of consciousness are attributable to the 5-HT2A receptor. *Elife*, *7*, e35082. https://doi.org/10.7554/eLife.35082
Preller, K. H., Razi, A., Zeidman, P., Stämpfli, P., Friston, K. J., & Vollenweider, F. X. 2019. "Effective connectivity changes in LSD-Induced altered states of consciousness in humans." *Proceedings of the National Academy of Sciences* 116 (7): 2743–2748.
Preller, K. H., & Vollenweider, F. X. (2016). Phenomenology, structure, and dynamic of psychedelic States In A. L. Halberstadt, F. X. Vollenweider, & D. E. Nichols (Eds.), *Behavioral neurobiology of psychedelic drugs* (Vol. 36, pp. 221–256). Springer.
Rao, R. P., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, *2*(1), 79. https://doi.org/10.1038/4580
Riba, J., Anderer, P., Morte, A., Urbano, G., Jané, F., Saletu, B., & Barbanoj, M. J. (2002). Topographic pharmaco-EEG mapping of the effects of the South American psychoactive beverage ayahuasca in healthy volunteers. *British Journal of Clinical Pharmacology*, *53*(6), 613–628. https://doi.org/10.1046/j.1365-2125.2002.01609.x
Riba, J., Romero, S., Grasa, E., Mena, E., Carrió, I., & Barbanoj, M. J. (2006). Increased frontal and paralimbic activation following Ayahuasca, the Pan-Amazonian inebriant. *Psychopharmacology*, *186*(1), 93–98. https://doi.org/10.1007/s00213-006-0358-7
Roseman, L., Haijen, E., Idialu-Ikato, K., Kaelen, M., Watts, R., & Carhart-Harris, R. (2019). Emotional breakthrough and psychedelics: Validation of the emotional breakthrough inventory. *Journal of Psychopharmacology*, *33*(9), 1076–1087. https://doi.org/10.1177/0269881119855974
Roseman, L., Leech, R., Feilding, A., Nutt, D. J., & Carhart-Harris, R. L. (2014). The effects of psilocybin and MDMA on between-network resting state functional connectivity in healthy volunteers. *Frontiers in Human Neuroscience*, *8*, 204. https://doi.org/10.3389/fnhum.2014.00204
Roseman, L., Nutt, D. J., & Carhart-Harris, R. L. (2018). Quality of acute psychedelic experience predicts therapeutic efficacy of psilocybin for treatment-resistant depression. *Frontiers in Pharmacology*, *8*, 974. https://doi.org/10.3389/fphar.2017.00974
Sherman, S. M., & Guillery, R. W. (2006). *Exploring the thalamus and its role in cortical function*. MIT press.
Strassman, R. J. (1995). Human psychopharmacology of N, N-dimethyltryptamine. *Behavioural Brain Research*, *73*(1–2), 121–124. https://doi.org/10.1016/0166-4328(96)00081-2

Strassman, R. J., Qualls, C. R., Uhlenhuth, E. H., & Kellner, R. (1994). Dose-response study of N, N-dimethyltryptamine in humans: II. Subjective effects and preliminary results of a new rating scale. *Archives of General Psychiatry*, *51*(2), 98–108. https://doi.org/10.1001/archpsyc.1994.03950020022002

Szocik, K., & Van Eyghen, H. (2021). *Revising cognitive and evolutionary science of religion. Religion as an adaptation*. Springer.

Tagliazucchi, E., Carhart-Harris, R., Leech, R., Nutt, D., & Chialvo, D. R. (2014). Enhanced repertoire of brain dynamical states during the psychedelic experience. *Human Brain Mapping*, *35*(11), 5442–5456. https://doi.org/10.1002/hbm.22562

Timmermann, C., Roseman, L., Schartner, M., Milliere, R., Williams, L. T., Erritzoe, D., Muthukumaraswamy, S., Ashton, M., Bendrioua, A., Kaur, O., Turton, S., Nour, M. M., Day, C. M., Leech, R., Nutt, D. J., & Carhart-Harris, R. L. (2019). Neural correlates of the DMT experience assessed with multivariate EEG. *Scientific Reports*, *9*(1), 1–13. https://doi.org/10.1038/s41598-019-51974-4

Tramacchi, D. T. (2006). Vapours and visions religious dimensions of DMT use.

Van Eyghen, H. (forthcoming). The brain perceives/Infers. A reply to vinten. In R. Vinten (Ed.), *Wittgenstein and the cgonitive science of religion*. Bloomsbury Academic.

Vollenweider, F. X., Leenders, K. L., Christian Scharfetter, P. M., Stadelmann, O., & Angst, J. (1997). Positron emission tomography and fluorodeoxyglucose studies of metabolic hyperfrontality and psychopathology in the psilocybin model of psychosis. *Neuropsychopharmacology*, *16*(5), 357–372. https://doi.org/10.1016/S0893-133X(96)00246-1

White, N. E., & Richards, L. M. (2009). *Introduction to quantitative EEG and neurofeedback: Advanced theory and applications* (T.H. Budzynski, J. R. Evans, & A. Abarbanel Eds.; 2nd ed., pp. 143–168). Academic Press.

Yaden, D. B., & Griffiths, R. R. (2020). The subjective effects of psychedelics are necessary for their enduring therapeutic effects. *ACS Pharmacology & Translational Science*, *4*(2), 568–572. https://doi.org/10.1021/acsptsci.0c00194

Yang, Y., & Raine, A. (2009). Prefrontal structural and functional brain imaging findings in antisocial, violent, and psychopathic individuals: A meta-analysis. *Psychiatry Research: Neuroimaging*, *174*(2), 81–88. https://doi.org/10.1016/j.pscychresns.2009.03.012